Lecture 05(b): The Lemke-Howson algorithm for general-sum normal-form games

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Outline of MAAI

1. Basic Game Theory and Nash Equilibrium
2. Potential Games and Best Respond Dynamics
3. Repeated Games and the Theory of Cooperation
4. Adversarial and Minimax Games
5. Bi-level Games and General-Sum Games and L-H Algorithms
6. Single-agent Reinforcement Learning and MDP
7. Learning Stochastic Games
8. Non-regret Learning and Correlated Equilibrium
9. Counterfactual Regret Minimisation
10. Learning with a Population of Agents

- Subject to change

Key References

- Bernhard Von Stengel. "Computing equilibria for two-person games". In: *Handbook of game theory with economic applications* 3 (2002), pp. 1723–1759
- Carlton E. Lemke and J. T. Howson. "Equilibrium points of bimatrix games". In: *SIAM Journal on Applied Mathematics* 12.2 (1964), pp. 413–423
- Olvi L Mangasarian and H Stone. "Two-person nonzero-sum games and quadratic programming". In: *Journal of Mathematical Analysis and applications* 9.3 (1964), pp. 348–355
- Lecture 2 - Zero-Sum Games; Minimax Theorem via Linear Programming: `https://stellar.mit.edu/S/course/6/sp17/6.853/courseMaterial/topics/topic3/lectureNotes/lec2/lec2.pdf`
- Game Theory and Algorithms Lecture 6: The Lemke–Howson Algorithm: `http://ints.io/daveagp/gta/lecture6.pdf`

Outline

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Computing NE of Two-Player General-Sum Games[1]

- Finding a Nash equilibrium of a two-player general-sum game cannot be formulated as a linear program,
- because the players' interests are not completely opposed.
- However, it can be framed as an optimisation problem.
- Next, we focus on bimatrix games, a type of normal-form game with two players:
  - Payoffs are represented by matrices $A, B \in \mathbb{R}^{m \times n}$.
  - Player 1 has m pure strategies and a mixed strategy $x \in \mathbb{R}^m$, where $x \geq 0$ and $\sum_{i=1}^{m} x_i = 1$.
  - Player 2 has n pure strategies and a mixed strategy $y \in \mathbb{R}^n$, where $y \geq 0$ and $\sum_{j=1}^{n} y_j = 1$.

[1] Bernhard Von Stengel. "Computing equilibria for two-person games". In: *Handbook of game theory with economic applications* 3 (2002), pp. 1723–1759.

Bilinear Programme for Nash Equilibria (NE)

- Solving a bimatrix game can be reduced to a bilinear programming problem[2].
- A mixed strategy profile (x^*, y^*) is a mixed Nash equilibrium of the bimatrix game (A, B) if and only if there exist auxiliary variables (p^*, q^*) such that (x^*, y^*, p^*, q^*) solves:

$$\begin{aligned} \text{maximise } & x^T A y + x^T B y - p - q \\ \text{subject to } & Ay \leq p\mathbf{1}_n, \\ & B^T x \leq q\mathbf{1}_m, \\ & \sum_i x_i = 1,\ \sum_j y_j = 1, x \geq 0,\ \ y \geq 0, \end{aligned}$$

where $\mathbf{1}_n$ ($\mathbf{1}_m$) is an n (m)-dimensional vector of ones. Here, p^* and q^* denote the maximum payoffs each player can guarantee against the opponent's strategy.

[2] Olvi L Mangasarian and H Stone. "Two-person nonzero-sum games and quadratic programming". In: *Journal of Mathematical Analysis and applications* 9.3 (1964), pp. 348–355.

Bilinear Programme for Nash Equilibria (NE): Insights

- For any feasible mixed strategies x, y:
  - Objective value $x^T Ay + x^T By - p - q \leq 0$.
- Any mixed strategy combination provides payoffs bounded by p and q
  - Payoffs for player 1: $x^T Ay \leq p$, and for player 2: $x^T By \leq q$.

Bilinear Programme for Nash Equilibria (NE): Insights[3]

- At Nash equilibrium (x^*, y^*):
  - Objective value equals zero: ${x^*}^T A y^* + {x^*}^T B y^* - p^* - q^* = 0$.
  - Players play best responses: $p^* = {x^*}^T A y^*$, $q^* = {x^*}^T B y^*$.
  - Thus Nash equilibria maximise the objective function, achieving zero objective value.

- Nash equilibria are guaranteed to exist and satisfy:

$$
\begin{aligned}
x^T A y^* &\leq p^*, \quad {x^*}^T A y^* = p^*, \\
{x^*}^T B y &\leq q^*, \quad {x^*}^T B y^* = q^*.
\end{aligned}
$$

The optimal value of the bilinear formulation is zero since it is non-positive and attained by the best response (therefore, a Nash equilibrium).

[3] Miriam Fischer and Akshay Gupte. "Multilinear formulations for computing Nash equilibrium of multi-player matrix games". In: *arXiv preprint arXiv:2208.03406* (2022).

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Computing NE of two-player general-sum games

- Alternatively, finding a Nash equilibrium of a two-player, general-sum game can be formulated as a linear complementarity problem (LCP)[4].
- The LCP will have no objective function at all.
  - It is a constraint satisfaction problem, or a feasibility programme, rather than an optimisation problem.
- Also, unlike zero-sum, we can no longer determine one player's equilibrium strategy by only considering other player's payoff (check the previous slides on zero-sum LP formulation);
  - Instead, need to discuss both players explicitly.

[4] Richard W Cottle and George B Dantzig. "Complementary pivot theory of mathematical programming". In: *Linear algebra and its applications* 1.1 (1968), pp. 103–125.

Linear Complementarity Problem for Computing NE[5]

- The LCP for computing the Nash equilibrium of a general-sum two-player game follows:

$$\sum_{k \in A_2} u_1(a_1^i, a_2^k) s_2^k + r_1^i = U_1^*, \quad \forall i \in A_1,$$
$$\sum_{j \in A_1} u_2(a_1^j, a_2^k) s_1^j + r_2^k = U_2^*, \quad \forall k \in A_2,$$
$$\sum_{j \in A_1} s_1^j = 1, \quad \sum_{k \in A_2} s_2^k = 1,$$
$$s_1^j \geq 0,\ s_2^k \geq 0, \quad \forall j \in A_1,\ \forall k \in A_2,$$
$$r_1^i \geq 0,\ r_2^k \geq 0, \quad \forall i \in A_1,\ \forall k \in A_2,$$
$$r_1^i s_1^i = 0,\ r_2^k s_2^k = 0, \quad \forall i \in A_1,\ \forall k \in A_2.$$

[5] Richard W Cottle and George B Dantzig. "Complementary pivot theory of mathematical programming". In: *Linear algebra and its applications* 1.1 (1968), pp. 103–125.

Linear Complementarity Problem for Computing NE

- Note 1: This formulation resembles strongly the LP formulation with slack variables, but without objective function.

$$\sum_{k \in A_2} u_1(a_1^i, a_2^k) s_2^k + r_1^i = U_1^*, \quad \forall i \in A_1,$$

$$\sum_{j \in A_1} u_2(a_1^j, a_2^k) s_1^j + r_2^k = U_2^*, \quad \forall k \in A_2,$$

$$\sum_{j \in A_1} s_1^j = 1, \quad \sum_{k \in A_2} s_2^k = 1,$$

$$s_1^j \geq 0, \; s_2^k \geq 0, \quad \forall j \in A_1, \; \forall k \in A_2,$$

$$r_1^i \geq 0, \; r_2^k \geq 0, \quad \forall i \in A_1, \; \forall k \in A_2,$$

$$r_1^i s_1^i = 0, \; r_2^k s_2^k = 0, \quad \forall i \in A_1, \; \forall k \in A_2.$$

Linear Complementarity Problem for Computing NE

- Note 2: The first constraint mirrors that in our LP formulation with slack variables; additionally, we include a second constraint to similarly restrict player 1's actions.

$$\sum_{k \in A_2} u_1(a_1^i, a_2^k) s_2^k + r_1^i = U_1^*, \quad \forall i \in A_1,$$

$$\sum_{j \in A_1} u_2(a_1^j, a_2^k) s_1^j + r_2^k = U_2^*, \quad \forall k \in A_2,$$

$$\sum_{j \in A_1} s_1^j = 1, \quad \sum_{k \in A_2} s_2^k = 1,$$
$$s_1^j \geq 0, \; s_2^k \geq 0, \quad \forall j \in A_1, \; \forall k \in A_2,$$
$$r_1^i \geq 0, \; r_2^k \geq 0, \quad \forall i \in A_1, \; \forall k \in A_2,$$
$$r_1^i s_1^i = 0, \; r_2^k s_2^k = 0, \quad \forall i \in A_1, \; \forall k \in A_2.$$

Linear Complementarity Problem for Computing NE

- Note 3: Also give the standard constraints that probabilities sum to one, that probabilities are nonnegative, and that slack variables are nonnegative, but now state these constraints for both players rather than only for player 1.

$$\sum_{k \in A_2} u_1(a_1^i, a_2^k) s_2^k + r_1^i = U_1^*, \quad \forall i \in A_1,$$

$$\sum_{j \in A_1} u_2(a_1^j, a_2^k) s_1^j + r_2^k = U_2^*, \quad \forall k \in A_2,$$

$$\sum_{j \in A_1} s_1^j = 1, \quad \sum_{k \in A_2} s_2^k = 1,$$

$$s_1^j \geq 0, \ s_2^k \geq 0, \quad \forall j \in A_1, \ \forall k \in A_2,$$

$$r_1^i \geq 0, \ r_2^k \geq 0, \quad \forall i \in A_1, \ \forall k \in A_2,$$

$$r_1^i s_1^i = 0, \ r_2^k s_2^k = 0, \quad \forall i \in A_1, \ \forall k \in A_2.$$

Linear Complementarity Problem for Computing NE

- Problem: With the constraints described so far, U_1^* and U_2^* would be allowed to take unboundedly large values, because all of these constraints remain satisfied when both U_1^* and r_1^i are increased by the same constant, for any given i and j.
- Solution: We solve this problem by adding the nonlinear constraint, called the complementarity condition.

$$r_1^j \geq 0, \quad s_1^j \geq 0, \quad r_1^j \cdot s_1^j = 0, \quad \forall j \in A_1$$

$$r_2^k \geq 0, \quad s_2^k \geq 0, \quad r_2^k \cdot s_2^k = 0, \quad \forall k \in A_2$$

The addition of this constraint means that we no longer have a linear programme; instead, we have a linear complementarity problem.

Solution: Complementary Slackness Condition

This constraint requires that:

- If a player assigns a positive probability to an action (in the support) in their mixed strategy, the corresponding slack variable is zero.
- Each slack variable indicates the player's incentive to deviate from that action.
- In equilibrium, actions played with positive probability yield equal expected payoffs; actions with lower expected payoffs are not played.
- Collectively, these conditions ensure each player plays a best response to the other's strategy, defining a Nash equilibrium.

$$r_1^j \geq 0, \quad s_1^j \geq 0, \quad r_1^j \cdot s_1^j = 0, \quad \forall j \in A_1$$
$$r_2^k \geq 0, \quad s_2^k \geq 0, \quad r_2^k \cdot s_2^k = 0, \quad \forall k \in A_2$$

Recall: Complementary Slackness Conditions (in LP)

- LP Strong Duality Theorem ($(cx^* = {y^*}^T A x^* = {y^*}^T b)$) gives:

$$(c - y^* A)x^* = 0 \quad (5)$$
$$y^*(b - Ax^*) = 0 \quad (6)$$

- which gives the following conditions:

$$x_j^* > 0 \implies y^* a_j = c_j, \quad y^* a_j > c_j \implies x_j^* = 0$$
$$y_i^* > 0 \implies a_i x^* = b_i, \quad a_i x^* < b_i \implies y_i^* = 0$$

This establishes a computable condition for a Nash Equilibrium.

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Best Response Condition

- Consider a general-sum matrix game (A, B) with strategies $x \in X$ (x as an m-vector) and $y \in Y$ (y as an n-vector).

Let

$$u = \max((Ay)_i | 1 \leq i \leq m),$$
$$v = \max((x^T B)_j | 1 \leq j \leq n).$$

Best Response Condition:

- x is a best response to y if and only if $x^T Ay = u$. Equivalently, we have:

$$x_i > 0 \implies (Ay)_i = u \quad \forall 1 \leq i \leq m.$$

- Similarly, y is a best response to x if and only if $x^T By = v$. This is equivalent to:

$$y_j > 0 \implies (x^T B)_j = v \quad \forall 1 \leq j \leq n.$$

Proof: Best Response Condition

$$x^T Ay = \sum_{i \in M} x_i (Ay)_i = \sum_{i \in M} x_i (u - (u - (Ay)_i))$$
$$= u - \sum_{i \in M} x_i (u - (Ay)_i).$$

So $x^T Ay \leq u$ because $x_i \geq 0$ and $u - (Ay)_i \geq 0$ for $1 \leq i \leq m$.

$$x^T Ay = u, x_i > 0 \implies (Ay)_i = u.$$

Thus, we also have:

- All pure strategies in the support of a Nash equilibrium strategy yield the same payoff.
- Definition: The support of a mixed strategy is the set of pure strategies that are played with non-zero probability ($x_i > 0$) under the mixed strategy.

Computing NE of Two-Player General-Sum Games

- We first consider an inner or totally mixed Nash equilibrium (x^*, y^*), i.e., $x_j^* > 0$ for all j (all pure strategies are played with positive probability).

Let

$a_i =$ the rows of payoff matrix A, $\quad b_i =$ the columns of payoff matrix B.

Using the fact that all pure strategies in the support of a Nash equilibrium strategy yield the same payoff, which is also greater than or equal to the payoffs for strategies outside the support, we have:

$$a_i^T y^* = u, \qquad \forall i = 1, 2, \ldots, n,$$
$$(x^*)^T b_j = v, \qquad \forall j = 1, 2, \ldots, m.$$

The preceding is a system of linear equations which can be solved efficiently.

Computing NE with Support Enumeration

- Most games lack totally mixed Nash equilibria, as not all strategies have positive probabilities.
- For a two-player game (A, B), a profile $(x, y) \in X \times Y$ is a Nash equilibrium with supports $S_1 \subseteq C_1$ and $S_2 \subseteq C_2$ if:

$$
\begin{aligned}
u &= a_i^T y, \quad && \forall i \in S_1, \quad & u &\geq a_i^T y, \quad && \forall i \notin S_1, \\
v &= x^T b_j, \quad && \forall j \in S_2, \quad & v &\geq x^T b_j, \quad && \forall j \notin S_2.
\end{aligned}
$$

(u, v) are Nash equilibrium values.

- Finding supports requires searching 2^{n+m} possibilities, leading to exponential complexity in the number of pure strategies.
- This complexity arises from determining the correct support sets for finite games.

Nondegenerate Games

Definition:

A two-player game is called nondegenerate if no mixed strategy of support size k has more than k pure best responses.

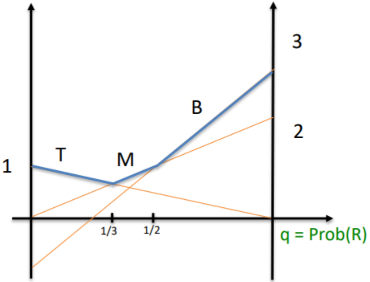
	L	R
T	1, 2	0, 0
M	0, 0	2, 1
B	-1, 4	3, 1

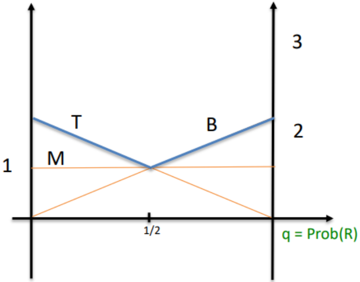
Nondegenerate games

	L	R
T	1, 0	0, 2
M	2, 1	3, 1
B	0, 2	2, 0

Degenerate games

Nondegenerate Games (Graphical Representation)





	L 1-q	R q
T	1, 2	0, 0
M	0, 0	2, 1
B	-1, 4	3, 1

Nondegenerate games

	L 1-q	R q
T	2, 0	0, 1
M	1, 3	1, 2
B	0, 2	2, 0

Degenerate games

Proposition and Proof

Proposition:

In any Nash equilibrium (x, y) of a nondegenerate two-player normal-form game, x and y have supports of equal size.

Proof: [Hint] Immediate from Best response condition.

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Support Enumeration Algorithm

Input: A nondegenerate two-player $m \times n$ matrix game
Output: All Nash equilibria of the game
Method:

- For each $k = 1, \ldots, \min\{m, n\}$:
  - Each pair (I, J) of k-sized subsets of m and n, respectively, solve the equations:

$$\sum_{i \in I} x_i b_{ij} = v \quad \text{for } j \in J, \qquad \sum_{i \in I} x_i = 1,$$
$$\sum_{j \in J} a_{ij} y_j = u \quad \text{for } i \in I, \qquad \sum_{j \in J} y_j = 1.$$

  - Check $x \geq 0, y \geq 0$, and the best response condition $\Rightarrow$ output x and y as Nash equilibrium.

Note: Linear equations may not have solutions. Nonunique solutions occur (for degenerate games).

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The Lemke–Howson Algorithm[6]

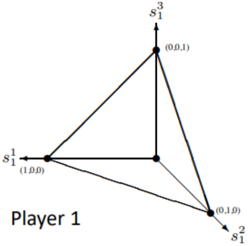
- The best-known algorithm to solve this LCP formulation is the Lemke–Howson algorithm.
- We will explain it through a graphical exposition.

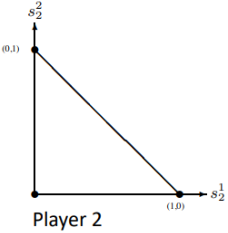
Player 2

Player 1

	$0, 1$	$6, 0$
	$2, 0$	$5, 2$
	$3, 4$	$3, 3$

Player 1 vs. Player 2





Each axis corresponds to one of the corresponding player's pure strategies, and the region spanned by these axes represents all the mixed strategies.

[6] Carlton E. Lemke and J. T. Howson. "Equilibrium points of bimatrix games". In: *SIAM Journal on Applied Mathematics* 12.2 (1964), pp. 413–423.

Labelling on the Strategies

- Every possible mixed strategy s_i is given a set of labels:

$$L(s_i) \subseteq A_1 \cup A_2,$$

  drawn from the set of available actions for both players.
- A given player as i and the other player as $-i$:
  - Mixed strategy s_i (as a whole) for player i is labelled as follows:
    - Label agent i's own actions a_i^j that is not in the support of s_i (with zero probability of choosing it), and
    - Label agent $-i$'s actions a_{-i}^j that is a best response by player $-i$ to s_i (could be more than one label).

Labelling on the Strategies: Implications

- This labelling is useful because a pair of strategies (s_1, s_2) is a Nash equilibrium if and only if it is completely labelled:

$$L(s_1) \cup L(s_2) = A_1 \cup A_2.$$

- For a pair to be completely labelled, each action a_i^j for agent i must:
  - (Labelled by agent i) either played by player i with zero probability, or
  - (Labelled by agent $-i$) be a best response by a_i^j to the mixed strategy of player $-i$. (but not both)

  So fulfil the best response condition for NE.

Labelling on the Strategies: Complementarity Condition

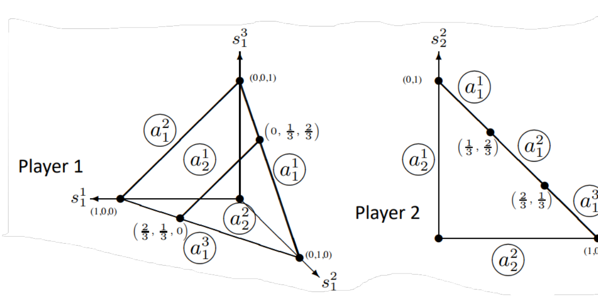
- This is a restatement of the complementarity condition given in constraint in our LCP:
  - Because the slack variable r_1^j is zero exactly when its corresponding action a_1^j is a best response to the mixed strategy s_{-i}.

$$\sum_{k \in A_2} u_1(a_1^j, a_2^k) \cdot s_2^k + r_1^j = U_1^* \quad \forall j \in A_1,$$

$$\sum_{j \in A_1} u_2(a_1^j, a_2^k) \cdot s_1^j + r_2^k = U_2^* \quad \forall k \in A_2.$$

  - if r_1^j is non-zero, then the corresponding s_1^j has to be zero (not in the support) according to the constraint $r_1^j \cdot s_1^j = 0$.

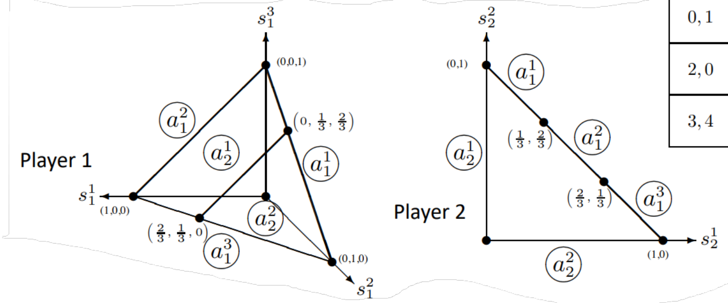
Labelled Strategy Space



$0, 1$	$6, 0$
$2, 0$	$5, 2$
$3, 4$	$3, 3$

- It is convenient to add one fictitious point in the strategy space of each agent, the origin; that is, $(0, 0, 0)$ for player 1 and $(0, 0)$ for player 2.
- Player 2's strategy space is a triangle with vertices $(0, 0), (1, 0), (0, 1)$, while player 1's strategy space is a pyramid with vertices $(0, 0, 0), (1, 0, 0), (0, 1, 0), (0, 0, 1)$.

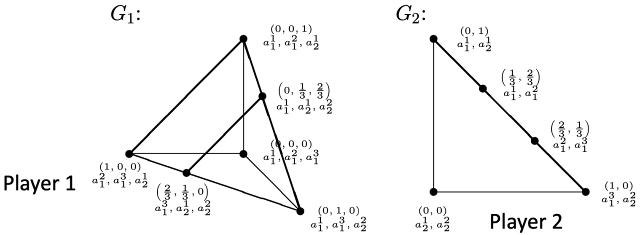
Labelled Strategy Space: Example



0, 1	6, 0
2, 0	5, 2
3, 4	3, 3

- Consider player 2:
  - The line from $(0, 0)$ to $(0, 1)$ is labelled with a_2^1 as none of them assign any probability to playing action a_2^1.
  - Similarly, the line from $(0, 0)$ to $(1, 0)$ is labelled with a_2^2.
  - Action a_1^1 is a best response by player 1 to any of the mixed strategies represented by the line from $(0, 1)$ to $(1/3, 2/3)$.
  - The point $(1/3, 2/3)$ is labelled by both a_1^1 and a_1^2, because both of these actions are best responses by player 1 to the mixed strategy $(1/3, 2/3)$ by player 2.

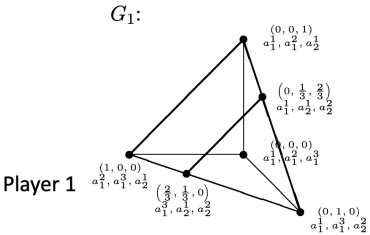
The Lemke–Howson Algorithm: Graphs G_1 and G_2

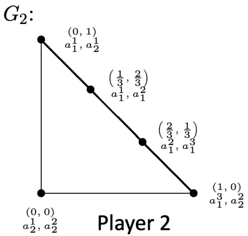


$0, 1$	$6, 0$
$2, 0$	$5, 2$
$3, 4$	$3, 3$

- Define G_1 and G_2 to be graphs for players 1 and 2 respectively.
- The nodes in the graph are fully labelled points in the labelled space:
  - Triply labelled points in G_1.
  - Doubly labelled points in G_2.
- An edge exists between pairs of points that differ in exactly one label.

The Lemke–Howson Algorithm: Graphs G_1 and G_2

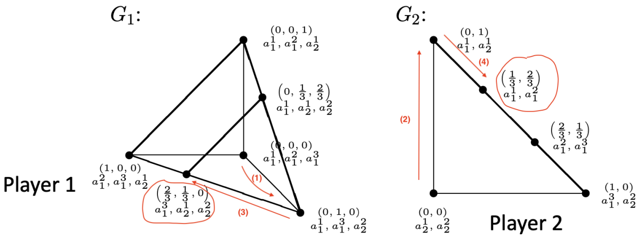




$0, 1$	$6, 0$
$2, 0$	$5, 2$
$3, 4$	$3, 3$

- The Lemke–Howson algorithm is to search these pairs of labelled spaces for a completely labelled pair (Nash equilibrium).
- In our example, the three Nash equilibria of the game:
  - $[(0, 0, 1), (1, 0)]$.
  - $[(0, 1/3, 2/3), (2/3, 1/3, 0)]$.
  - $[(2/3, 1/3, 0), (1/3, 2/3)]$.

The Lemke–Howson Algorithm: Execution Example



$0,1$	$6,0$
$2,0$	$5,2$
$3,4$	$3,3$

- A possible execution of the algorithm starts at $(0,\ 0)$ and:
  - (1) Changes s_1 to $(0,1,0)$. Now, our pair $(0,1,0),(0,0)$ is a_1^2-almost completely labelled, and the duplicate label is a_2^2.
  - (2) In G_2, moves to $(0,1)$ because the other possible choice $(1,0)$ has label a_2^2.
  - (3) Returns to G_1, moves to $(2/3,1/3,0)$ adjacent to $(0,1,0)$, removing duplicate a_1^1.
  - (4) Final step: change s_2 to $(1/3,2/3)$, avoiding a_2^1.
- Execution trace:
  $(0,0,0),(0,0)) \to ((0,1,0),(0,0)) \to ((0,1,0),(0,1)) \to ((2/3,1/3,0),(0,1)) \to ((2/3,1/3,0),(1/3,2/3))$.

Conclusions

- We have learned:
  - A simple solution for finding Nash Equilibrium.
  - The Minimax theorem and the Strong Duality theorem of LP are equivalent.
  - Linear programming solutions for finding Nash Equilibrium in zero-sum games.
  - The Lemke–Howson algorithm for finding sample Nash Equilibrium in general-sum games.

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References I

[CD68] Richard W Cottle and George B Dantzig. "Complementary pivot theory of mathematical programming". In: *Linear algebra and its applications* 1.1 (1968), pp. 103–125.

[FG22] Miriam Fischer and Akshay Gupte. "Multilinear formulations for computing Nash equilibrium of multi-player matrix games". In: *arXiv preprint arXiv:2208.03406* (2022).

[LH64] Carlton E. Lemke and J. T. Howson. "Equilibrium points of bimatrix games". In: *SIAM Journal on Applied Mathematics* 12.2 (1964), pp. 413–423.

[MS64] Olvi L Mangasarian and H Stone. "Two-person nonzero-sum games and quadratic programming". In: *Journal of Mathematical Analysis and applications* 9.3 (1964), pp. 348–355.

References II

[Von02] Bernhard Von Stengel. "Computing equilibria for two-person games". In: *Handbook of game theory with economic applications* 3 (2002), pp. 1723–1759.